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Exploring the Emergence of Social Order through Computational Social Science and SNS Data

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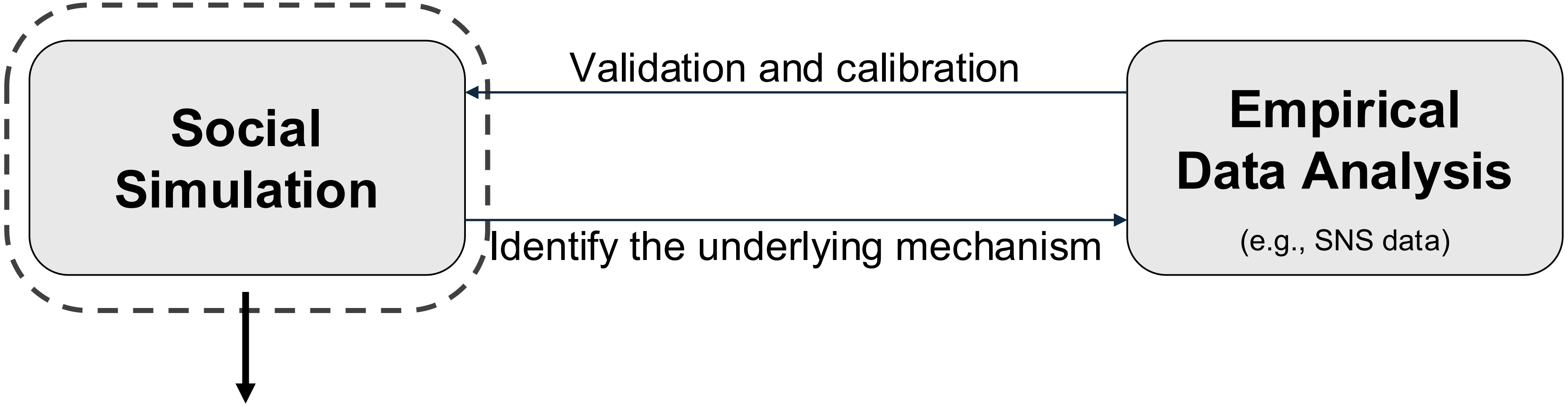
Research Framework

- **Aim**: Investigate the underlying mechanisms behind the emergence, maintenance, and evolution of social order.



This presentation focuses on
how to establish a social simulation model for investigating social order

Social order as a central research topic

- Social order is a necessary condition for society to exit.
- How does social order emerge?
- Thomas Hobbs
  - Individuals transfer their rights to a sovereign power to avoid the war of all against all.
- Adam Smith
  - Economic exchange among self-interested persons
- **Talcott Parsons**
  - **Mutual reinforcement between factual and normative orders**

Emergence of social order *à la* Parsons

- Social order: A process of mutual reinforcement between factual and normative orders leads to the emergence of social order
- **Factual order:** Regular behaviors emerge in society
  - E.g. During the COVID-19 pandemic, many people regularly wore masks.
- **Normative order:** People begin to believe that these regular behaviors are desirable and expect others to hold the same belief
  - E.g. Seeing others wear masks, individuals may come to think that mask-wearing is socially desirable and that others agree.

Limitations in conventional social simulation

- Often rely on **abstract theories** and **oversimplified assumptions** to define agents
  - Hard to define agents' behaviors in accordance with abstract concepts such as *normative order*.

- Real-world contexts are usually **conceptualized into abstract components**.
  - Loss of realism and nuance
  - Validation challenges
  - …

LLMs-driven Agent based Model

Use LLMs-driven Agent to replace agents in social simulation

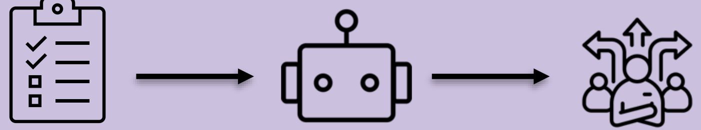
$$X \rightarrow f(X)$$

Pre-defined rules

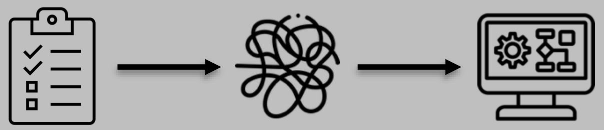
Definition
(in natural language)

LLM
Agent

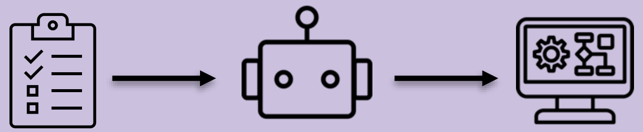
Behaviors
Decision-making



Sophisticated and human-like
behavior and decision-making
by LLMs



Real-world contexts are
conceptualized into abstract components



LLM incorporate context into the model
through natural language
enabling **simulations that more closely**
reflect real-world conditions

Research strategy

- **Examine the efficacy of of LLMs-driven agents in reproducing the emergence of social norm (Bicchieri, 2006).**
  - **Empirical expectations**: Beliefs about what others actually do.
  - **Normative expectations**: Beliefs about what others think one *should* do, and what they expect *you* to do.

- The combination of empirical and normative expectations is necessary for a stable social norm.
  - (In other words) social norm can not be established unless both the empirical expectations and normative expectations are in a high level.

Prompt for Agent Setting

You are a worker in a company. In your company, sometimes some workers choose to work overtime.

Some people think overtime work is acceptable since working overtime helps them meet deadlines and perform better in evaluations, and staying late signals dedication to the boss and may help them get promoted.
Some people think overtime work is unacceptable because work-life balance is important.

It’s almost the end of the workday, and you are trying to decide whether to work overtime today.
Your decision depends on the following three factors:

1. Your approval of overtime work (approval_score)is **2** (Scale: 1–10)
This reflects how much you personally believe working overtime is reasonable. A higher score means you think overtime is more justified.

2. Your perception of others’ expectations (other_expectations) is **3** (Scale: 1–10)
This reflects how much you believe other people (your boss, coworkers) expect you to stay late. A higher score means you believe that if you leave on time, others might see you as less dedicated.

3. Your observation of the workplace yesterday is **you observed 4 colleagues. Among them, 1 worked overtime.**
Your approval_score and other_expectations are personal and will change based on your observations.

Context
The emergence of “overwork” norm

Normative expectations

Empirical expectations

Prompt for Agent Setting

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### Please follow these steps to make your decision:

**Step 1: Observe the environment**
Take a moment to think about what you observe in the office yesterday. Are many
people staying late? What does this suggest to you about this workplace culture?
**Step 2: Reflect and adjust your mindset**
Based on your observations, update your internal scores:
- What is your updated **approval_score** (1–10)?
- What is your updated **other_expectations** score (1–10)?
Explain how your observation affected your perception. Why did you increase or
decrease the scores?
It should change gradually over time, as human beliefs typically do.

**Step 3: Make a decision**
Based on your updated approval_score, your perception of others' expectations, and
what you observed in your workplace, decide whether to stay and work overtime today.
Please explain your reasoning in natural language.
Finally, output your decision in the following format: ....
```

Observed situations influence **normative expectations** and **personal normative beliefs**.

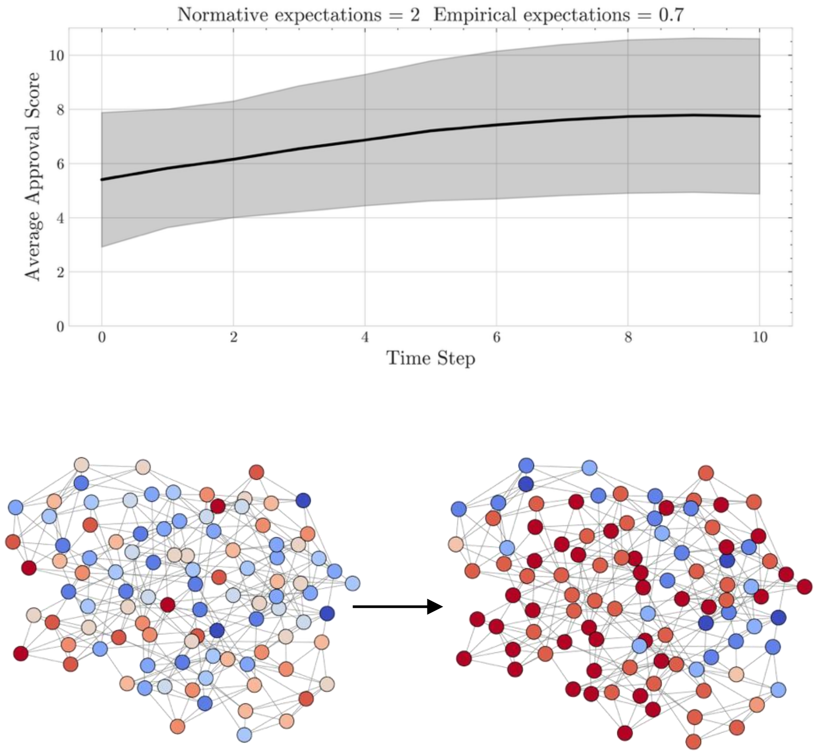
One's own actions, normative expectations, and personal normative beliefs are updated, and behavior is determined based on them.

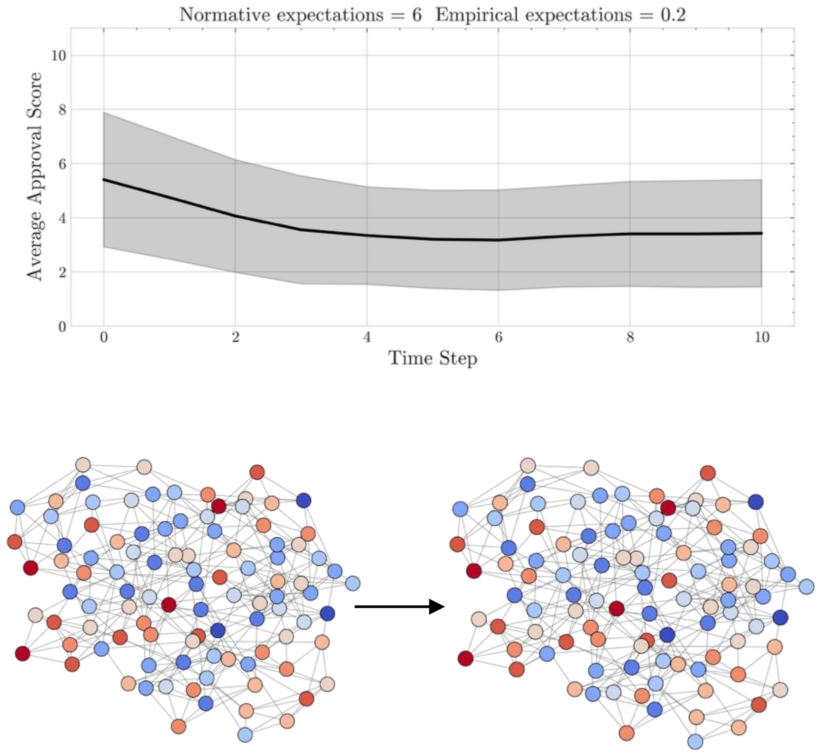
Simulation Setting

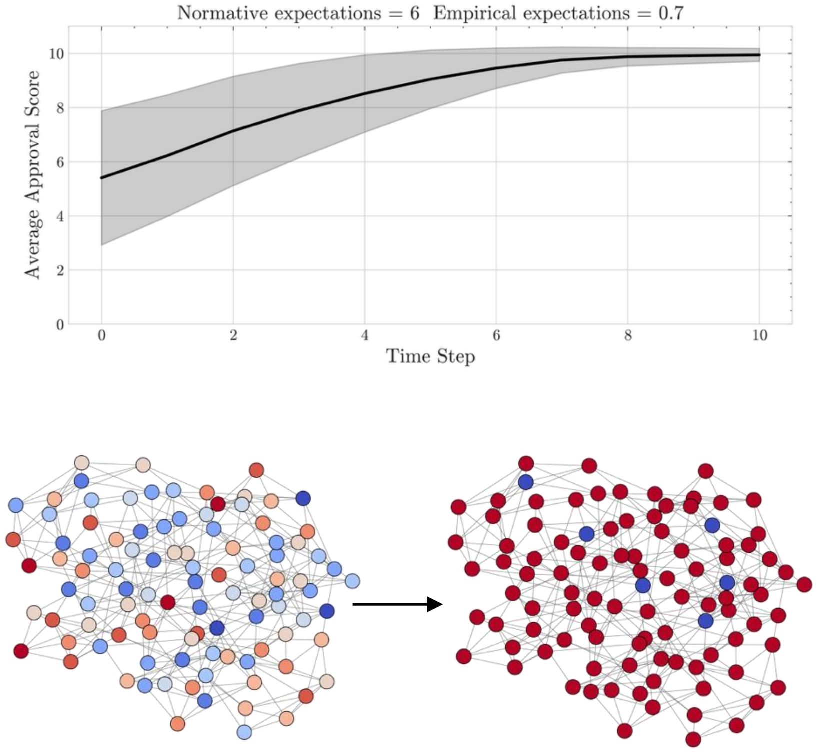
- **LLMs model**: Gemini 2.5 Flash
- **Total number of agents**: N=100
- **Environment Settings**: Small-world Network
- **Normative expectations** follow a normal distribution with a mean in $\in \{1,2,\ldots,7\}$ and standard deviation $\sigma = 2$
- The proportion of agents who engage in "overtime work" $\in \{0.1,0.2,\ldots,0.7\}$
  - The proportion affects the **empirical expectations**.
  - Since the norm is assumed to be not yet well-established at initial phase, the proportions are set at relatively modest levels.

Investigate the approvement to the "overwork" (= Social Norm)
in various scenarios
with different normative expectations and empirical expectations

Results





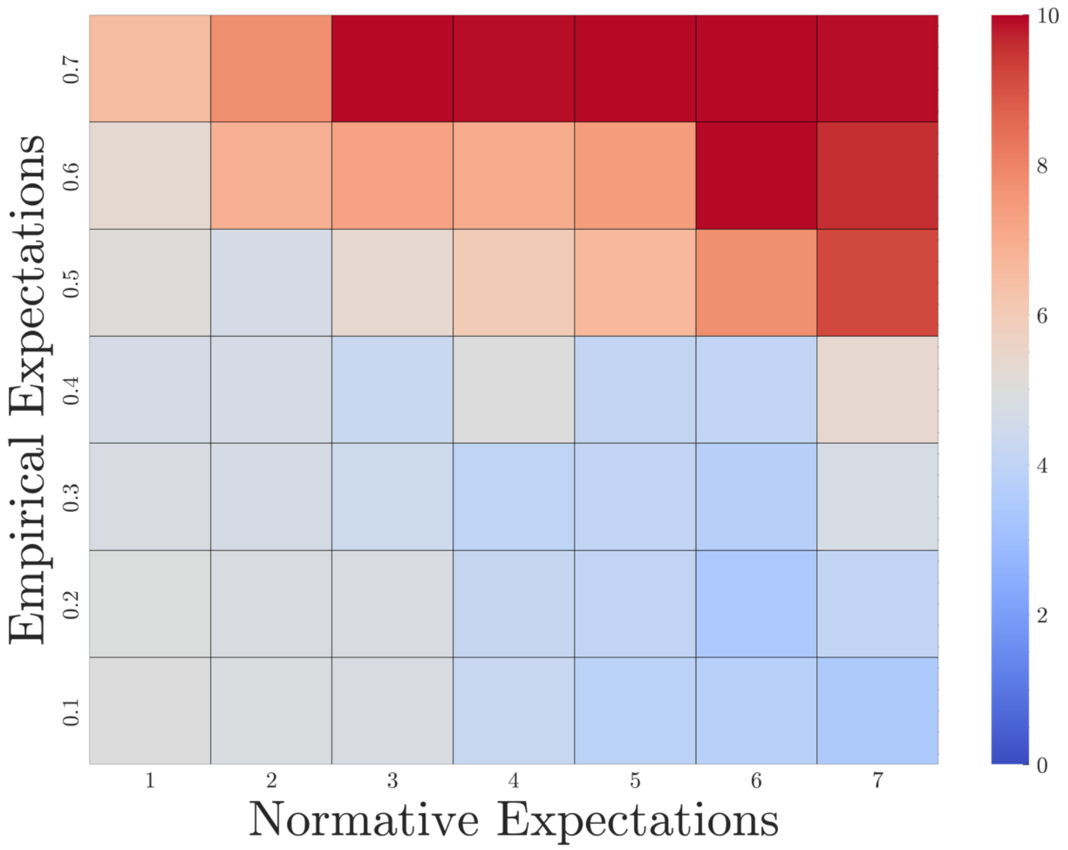


High EE/Low NE **Low EE/High NE** **High EE/High NE**

Results

- The average approvement to the social norm at the final step of the simulation across different scenarios.

- Social norm is more likely to be established when **empirical expectations and normative expectations are both in a high level**
  - Simulation reproduced the emergence of social norm indicated by Bicchieri (2006)



Conclusions

- LLM-based agents can efficiently construct social simulations to investigate the emergence of social norms.
  - Replicate the theoretica mechanisms underlying the emergence of social norms.
- High adaptability to diverse contexts and scenarios
  - adapted and repurposed for different social contexts, demographic groups, or hypothetical scenarios simply by modifingy the prompts
- Effective intergration with empirical analysis
  - Real-world context and insights from empirical analysis can be incorporated efficiently through natural language prompts